From Monocular Visual Odometry to Metric Stereo Visual Odometry on TUM VI

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***Abstract*—This paper presents a classical geometry-based visual odometry pipeline on the TUM VI benchmark. The project starts from a monocular visual odometry baseline and extends it to metric stereo visual odometry using calibrated stereo depth. The monocular pipeline estimates relative camera motion using feature matching, essential matrix estimation, pose recovery, and frame-to-frame pose chaining. The stereo pipeline uses fisheye stereo rectification, disparity-derived 3D points, and 3D–2D Perspective-n-Point estimation with RANSAC to recover metric motion. Experiments are performed on Room2, Corridor3, and Outdoors5. Room2 is evaluated using full ground truth with ATE and RPE, while Corridor3 and Outdoors5 are evaluated using start–end drift. On Room2, Stereo VO improves ATE RMSE from 1.217 m to 0.944 m and RPE RMSE from 0.945 m to 0.339 m. Corridor3 raw Stereo VO drift is reduced from 46.99 m to 6.75 m using two verified visual loop closures. Outdoors5 raw Stereo VO drift is reduced from 32.29 m to 18.31 m using a safe soft SIFT loop correction. The paper also discusses practical implementation problems, including fisheye rectification, disparity noise, unstable PnP, false loop closures, and trajectory over-correction.**

***Index Terms*—visual odometry, stereo visual odometry, monocular visual odometry, TUM VI, disparity, PnP, RANSAC, loop closure**

I. INTRODUCTION

Visual odometry (VO) estimates camera motion from image sequences and is an important component of robotics, autonomous navigation, augmented reality, and SLAM systems. A monocular camera can recover relative motion from image correspondences, but the translation scale is unknown. This scale ambiguity causes accumulated drift and makes monocular VO unsuitable for direct metric trajectory estimation without additional assumptions.

Stereo visual odometry reduces this limitation by using two synchronized cameras with known calibration and baseline. Disparity between the left and right images allows image points to be back-projected into metric 3D coordinates. Motion estimation can then be formulated as a 3D–2D pose-estimation problem instead of a scale-ambiguous monocular problem.

The objective of this project is to implement a classical progression from Monocular VO to Stereo VO on the TUM VI benchmark. The implementation uses only classical computer vision: feature matching, epipolar geometry, stereo rectification, disparity, PnP, RANSAC, and optional visual loop closure. No deep learning, neural depth, or learned features are used.

The main contributions are:

- a Monocular VO baseline using essential matrix pose recovery;
- a metric Stereo VO pipeline using disparity and PnP-RANSAC;
- evaluation on Room2, Corridor3, and Outdoors5;
- practical failure analysis and debugging discussion;
- optional visual loop-closure improvements for long sequences.

II. RELATED WORK

Classical visual odometry has been widely studied in robotics and computer vision. Scaramuzza and Fraundorfer provide a detailed overview of feature-based VO and explain the main sources of drift in practical systems [1]. Hartley and Zisserman describe the multiple-view geometry foundations used in this project, including essential matrices, stereo geometry, triangulation, and camera pose recovery [2].

ORB-SLAM2 shows how classical feature-based methods can be extended into a complete SLAM system using keyframes, map points, local bundle adjustment, relocalization, and loop closure [3]. The present project is intentionally simpler. It focuses on the front-end visual odometry pipeline and does not include full SLAM optimization. Therefore, larger drift is expected in long sequences.

The TUM VI benchmark provides synchronized stereo images, calibration, IMU data, and ground truth for visual-inertial odometry evaluation [4]. In this project, only stereo images and calibration are used. IMU measurements are not integrated, so the system remains a pure visual odometry pipeline.

III. DATASET AND EXPERIMENTAL SETUP

A. Sequences

The experiments use three required TUM VI sequences:

- **Room2**: indoor room sequence with full ground truth;
- **Corridor3**: indoor corridor sequence evaluated by start–end drift;
- **Outdoors5**: long outdoor sequence evaluated by start–end drift.

The images have resolution 512×512. The experiments use approximately 2882 frames for Room2, 5802 frames for Corridor3, and 17747 frames for Outdoors5.

B. Calibration and Repository Setup

The official Kalibr-style `camchain.yaml` file is used for camera intrinsics, distortion, and stereo extrinsics. Since TUM VI uses fisheye/equidistant cameras, direct pinhole stereo matching is not valid. Therefore, stereo rectification is performed before feature matching and disparity computation. The stereo baseline is approximately 0.101 m. The final implementation, notebooks, output tables, plots, and trajectory files are publicly available in the project repository [6].

IV. Methodology

A. Pipeline Overview

The full pipeline is shown conceptually as:

$$\text{Images} \rightarrow \text{Rectification} \rightarrow \text{Features/Disparity}$$
$$\rightarrow \text{Motion Estimation} \rightarrow \text{Trajectory}$$

The monocular and stereo pipelines share the same goal of estimating frame-to-frame camera motion, but they differ in scale observability. Monocular VO estimates motion from 2D–2D geometry, while Stereo VO estimates motion from metric 3D–2D correspondences.

B. Monocular Visual Odometry

The monocular pipeline uses only the left camera. Features are detected and matched between consecutive frames. Given normalized corresponding image points $\mathbf{x}$ and $\mathbf{x}'$, the essential matrix satisfies:

$$\mathbf{x}'^T \mathbf{E} \mathbf{x} = 0. \tag{1}$$

The essential matrix can be decomposed as:

$$\mathbf{E} = [\mathbf{t}]_\times \mathbf{R}, \tag{2}$$

where $\mathbf{R}$ is relative rotation and $\mathbf{t}$ is translation direction. RANSAC is used to reject outliers, and the physically valid pose is selected using cheirality. The relative poses are chained to form a trajectory. Since monocular VO cannot recover absolute scale, Sim(3) alignment is used for Room2 evaluation.

C. Stereo Visual Odometry

The stereo pipeline uses rectified stereo images to compute disparity. For a point with disparity d, depth is:

$$Z = \frac{fB}{d}, \tag{3}$$

where f is focal length and B is the stereo baseline.

The corresponding 3D point is:

$$\mathbf{X} = \begin{bmatrix} (u - c_x)Z/f_x \\ (v - c_y)Z/f_y \\ Z \end{bmatrix}. \tag{4}$$

These 3D points from the previous frame are matched or tracked to 2D observations in the current left image. Motion is then estimated using PnP with RANSAC. This produces a metric relative pose, which is chained into a global trajectory.

D. Robust Filtering

Several filtering steps are required for stability:

- reject invalid or very small disparities;
- reject far-depth points with unstable triangulation;
- filter weak feature matches and geometric outliers;
- reject PnP estimates with too few inliers;
- skip unrealistic translation or rotation steps;
- reject high reprojection-error solutions.

These filters are important because small frame-to-frame errors accumulate over long sequences.

TABLE I
MAIN IMPLEMENTATION SETTINGS

Component	Setting
Image size	512×512
Stereo baseline	≈ 0.101 m
Feature matching	ORB / SIFT experiments
Disparity	StereoSGBM
Motion model	3D–2D PnP
Outlier rejection	RANSAC
Room2 alignment	Sim(3) mono, SE(3) stereo
Corridor3/Outdoors5 metric	Start–end drift
Loop validation	Visual + geometric + PnP checks

V. Loop Closure Extensions

A. Corridor3

Corridor3 contains repeated corridor structures and long forward motion. Raw Stereo VO accumulated large drift. To reduce this, a visual loop search was implemented using feature matching, Fundamental-matrix verification, and dense stereo-PnP validation. Two reliable loops were selected:

- frame $560 \rightarrow 4760$;
- frame $780 \rightarrow 5060$.

The loop corrections were applied as trajectory position corrections. Ground truth was not used for loop detection or correction; it was used only for final evaluation.

B. Outdoors5

Outdoors5 was more challenging for loop closure. A relaxed ORB loop produced a very low numerical drift, but the visual match evidence was incorrect, so it was rejected as a false loop closure. A stricter ORB/PnP search found no reliable loop. To handle viewpoint rotation and outdoor texture, a SIFT rotation-compensated loop search was used.

A valid loop was found at:

$$6000 \rightarrow 11847. \tag{5}$$

Full correction over-deformed the trajectory because the accumulated drift was large. Therefore, a soft correction with gain 0.5 was used. This reduced drift while avoiding the strongest over-correction.

VI. Evaluation Protocol

Room2 has full ground truth, so ATE and RPE are reported. Monocular VO uses Sim(3) alignment because of scale ambiguity, while Stereo VO uses SE(3) alignment because it is metric.

ATE RMSE is:

$$\text{ATE}_{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} \left\| \mathbf{p}_i - \mathbf{p}_i^{gt} \right\|^2}. \tag{6}$$

For Corridor3 and Outdoors5, start–end drift is used:

$$\epsilon_{\text{drift}} = \left\| (\mathbf{p}_{end} - \mathbf{p}_{start}) - (\mathbf{p}_{end}^{gt} - \mathbf{p}_{start}^{gt}) \right\|. \tag{7}$$

All final outputs were saved in a reproducible structure containing four folders: `configs/`, `trajectories/`, `tables/`, and `plots/`. Trajectories were exported in TUM format:

$$timestamp,\ t_x,\ t_y,\ t_z,\ q_x,\ q_y,\ q_z,\ q_w. \tag{8}$$

This allows the results to be checked independently from the notebooks.

VII. Results and Analysis

A. Room2 Full Ground-Truth Evaluation

TABLE II
Room2 Full Ground-Truth Evaluation

Method	Align.	ATE RMSE	RPE RMSE
Monocular VO	Sim(3)	1.217 m	0.945 m
Stereo VO	SE(3)	0.944 m	0.339 m

Stereo VO improves both ATE and RPE. This confirms that calibrated stereo depth reduces scale ambiguity and improves local motion consistency.

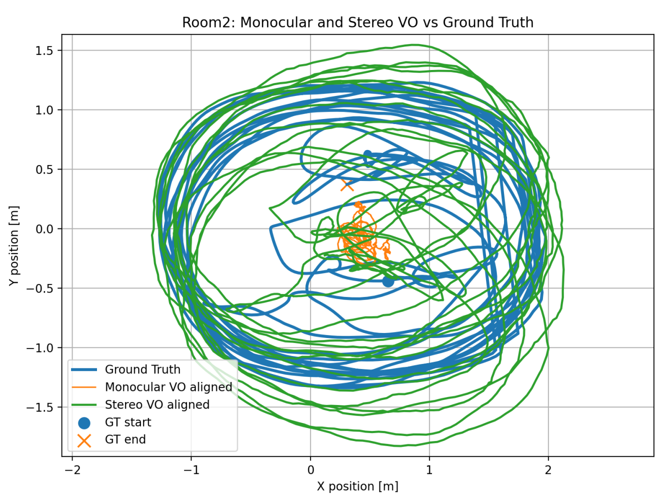

Fig. 1. Room2 trajectory comparison. Monocular VO is Sim(3)-aligned and Stereo VO is SE(3)-aligned.

TABLE III
Corridor3 Start–End Drift

Method	Drift
Raw Stereo VO	46.99 m
Stereo VO + two visual loops	6.75 m

B. Corridor3 Start–End Drift

The large raw drift shows the limitation of frame-to-frame Stereo VO in repeated indoor structures. The two verified loops significantly reduce the final endpoint error.

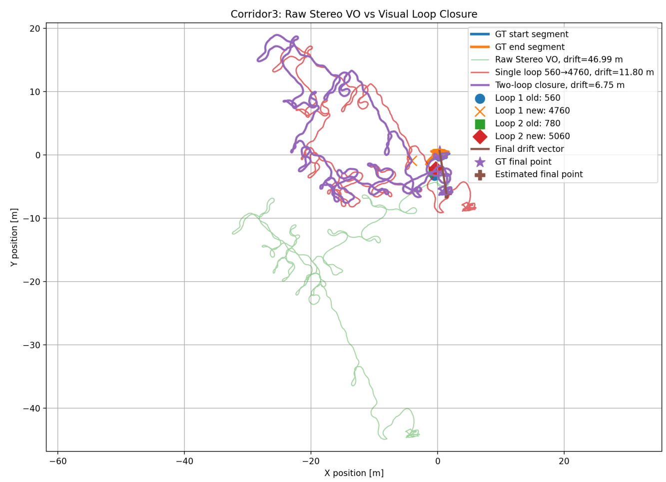

Fig. 2. Corridor3 raw Stereo VO and loop-corrected trajectory. Two verified loops reduce accumulated drift.

C. Outdoors5 Start–End Drift

TABLE IV
Outdoors5 Start–End Drift

Method	Drift
Monocular VO baseline	2.41 m
Raw Stereo VO baseline	32.29 m
Stereo VO + step smoothing	29.87 m
Stereo VO + safe soft SIFT loop	18.31 m

The Monocular VO result is numerically small after Sim(3) alignment, but it is not a metric stereo trajectory. The raw Stereo VO is metric but accumulates drift over the long sequence. The safe soft SIFT loop reduces drift while avoiding the false-loop and over-correction problems.

D. Final Summary

TABLE V
Final Results Summary

Dataset	Method	Result
Room2	Monocular	ATE 1.217, RPE 0.945
Room2	Stereo	ATE 0.944, RPE 0.339
Corridor3	Raw Stereo	46.99 m drift
Corridor3	Stereo + loops	6.75 m drift
Outdoors5	Raw Stereo	32.29 m drift
Outdoors5	Stereo + soft loop	18.31 m drift

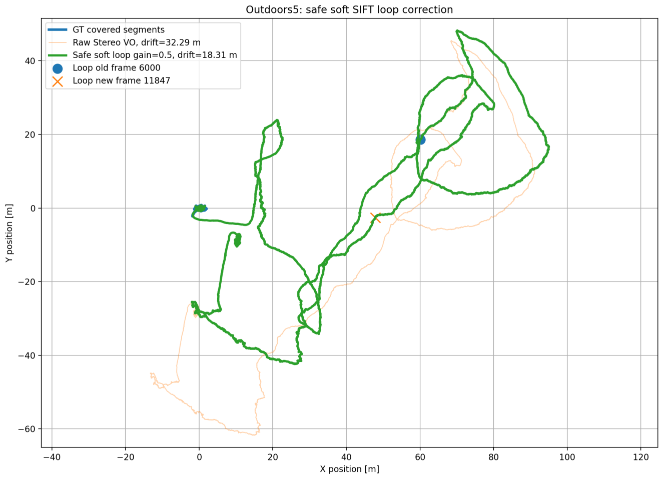

Fig. 3. Outdoors5 raw Stereo VO, full SIFT loop, and safe soft SIFT loop correction.

E. Runtime Summary

Runtime was recorded to show the computational cost of the final VO pipelines. Monocular VO is generally faster because it estimates motion from 2D–2D image correspondences, while Stereo VO additionally performs stereo rectification, disparity/depth processing, 3D–2D correspondence construction, and PnP-RANSAC. The optional loop-correction steps are treated as post-processing and are not included in the raw VO runtime.

TABLE VI
RUNTIME SUMMARY OF FINAL RUNS

Dataset	Method	Frames	Runtime
Room2	Monocular VO	2882	10.15 s
Room2	Stereo VO	2882	11.84 s
Corridor3	Monocular VO	5802	154.96 s
Corridor3	Stereo VO	5802	284.05 s
Outdoors5	Monocular VO	17747	604.98 s
Outdoors5	Stereo VO	17747	1152.51 s

Loop-corrected results are post-processing steps applied after the raw Stereo VO trajectory, so their cost is not included in the raw VO runtime.

VIII. PROBLEMS ENCOUNTERED AND FIXES

A. Fisheye Rectification

An early problem was trying to use the raw fisheye stereo images directly. This produced unreliable disparity and poor matching. The fix was to use the official Kalibr calibration and apply fisheye stereo rectification before feature matching and disparity computation.

B. Disparity Noise

Disparity was noisy in weak-texture and far-depth areas. This produced incorrect 3D points and unstable PnP. The fix was to reject small disparities, clamp far depth, and use local patch-based disparity validation.

C. PnP Instability

Some frame pairs produced too few 3D–2D correspondences or too few RANSAC inliers. Integrating these updates caused large pose jumps. The fix was to enforce minimum correspondence counts, minimum PnP inliers, reprojection RMSE thresholds, translation limits, and rotation limits.

D. False Loop Closure

A relaxed Outdoors5 loop produced an unrealistically low drift, but visual evidence showed that the images were not the same place. It was rejected. This showed that loop closure must be validated visually and geometrically, not only by final error.

E. Loop Over-Correction

A full SIFT loop correction on Outdoors5 improved the metric but bent the trajectory strongly. The fix was a soft loop gain of 0.5. This is a conservative pose-graph-inspired correction that improves drift without forcing the whole trajectory to one loop constraint.

IX. DISCUSSION

The experiments show that stereo depth improves metric consistency, especially on Room2. However, long sequences reveal the limitations of pure frame-to-frame VO. Without bundle adjustment, keyframes, map reuse, relocalization, or global pose graph optimization, even metric Stereo VO accumulates drift.

Corridor3 demonstrates the importance of loop closure in repeated indoor structures. Outdoors5 demonstrates the danger of false loop closures: a candidate can improve the numerical metric but still be geometrically wrong. Therefore, visual evidence and stereo-PnP validation are essential.

The results also show that Monocular VO can sometimes look strong after Sim(3) alignment, especially in start–end drift. However, this does not mean it is metric. Stereo VO remains the method that estimates real-scale motion from the calibrated baseline.

A. Limitations

The implemented system is a visual odometry pipeline, not a full SLAM system. It does not include keyframe-based mapping, local bundle adjustment, relocalization, or global pose-graph optimization. Therefore, long sequences can still accumulate drift even when stereo depth is available. The loop-closure experiments reduce this drift, but they also show that false loop candidates and over-correction must be handled carefully.

X. CONCLUSION

This project implemented a classical transition from Monocular VO to Stereo VO on TUM VI. The Monocular VO baseline demonstrated scale ambiguity, while Stereo VO used calibrated disparity and PnP to estimate metric motion.

On Room2, Stereo VO improved ATE RMSE from 1.217 m to 0.944 m and RPE RMSE from 0.945 m to 0.339 m. On

Corridor3, two verified visual loops reduced Stereo VO drift from 46.99 m to 6.75 m. On Outdoors5, a safe soft SIFT loop reduced raw Stereo VO drift from 32.29 m to 18.31 m.

The system is not a full SLAM system and does not include local bundle adjustment or global pose-graph optimization. Nevertheless, it satisfies the project goal by demonstrating the transition from up-to-scale monocular motion to metric stereo odometry and by honestly reporting the practical failures and fixes encountered during implementation.

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